PALEOMAGNETIC AND ROCK-MAGNETIC SURVEY OF EOCENE DIKE SWARMS FROM THE TECALITLAN AREA (WESTERN MEXICO): TECTONIC IMPLICATIONS

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ABSTRACT

For long time the western-central Mexico has been affected by oblique subduction caused by Farallon plate beneath North America. As result, smaller plates (e.g. Cocos Plate), several fault systems outlining crustal blocks (e.g. Michoacán block) and magmatic arcs (e.g. Paleocene-Early Oligocene magmatism and the Trans-Mexican Volcanic Belt) were developed. Still, no paleomagnetic data are available for Oligocene and Miocene. The principal aim of this study is to evaluate whether the tectonic rotations and relative motions of these blocks occurred before the Miocene. Here, we report a detailed rock-magnetic and paleomagnetic results from Tecalitlan area, located in the Michoacán block. Sixteen sites (about 150 oriented samples) were collected including one radiometrically dated diabase dike (35.0 ± 1.8 Ma). Rock-magnetic experiments permitted identification of magnetic carriers and assessment of the paleomagnetic stability. Continuous susceptibility measurements vs temperature in most cases yield reasonably reversible curves with Curie points close to that of magnetite. Reliable paleomagnetic directions were obtained for 12 sites. Inclination I and declination D of the mean paleomagnetic direction obtained in this study are I = 33.1°, D = 345.0°, and Fisherian statistical parameters are k = 25, α_{95} = 8.9°. The corresponding mean paleomagnetic pole position is P_{lat} = 75.7°, P_{long} = 166.6°, K = 31, A_{95} = 8.0°. The mean inclination is in

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reasonably good agreement with the expected value, as derived from reference poles for the stable North America. Magnetic declination is not significantly different from that expected which is in disagreement with a counterclockwise tectonic rotation of about 20º previously reported for the studied area. Based on paleomagnetic results obtained in this study compiled with those currently available from the Michoacán Block, we propose a simple model suggesting that sometime in Eocene epoch the convergence vector of the Farallon plate relative to North America plate was normal to the trench before reaching an actual oblique convergence.

Keywords: Western Mexico, paleomagnetism, dyke swarm, counterclockwise rotation

1. INTRODUCTION

Western Mexico has been affected by the subduction of the oceanic Farallon plate under North American plate (*Atwater, 1970*). This processes left behind smaller oceanic plates (e.g. Cocos and Rivera plates) and a complex geology including modern volcanic arcs from Paleocene to present day (e.g. the Sierra Madre Occidental and the Trans-

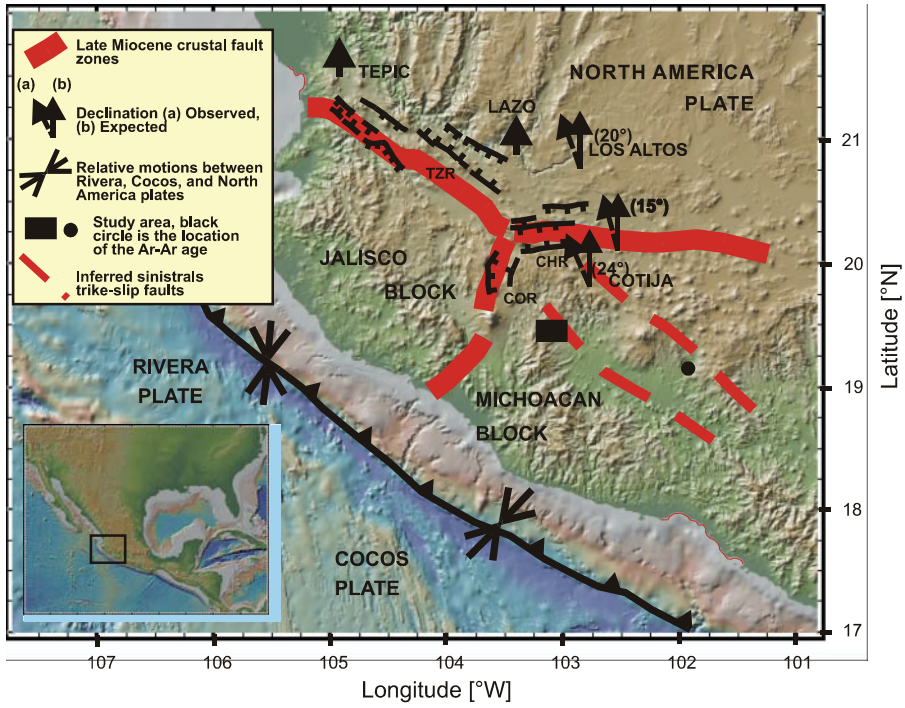


Fig. 1. Digital Elevation Model of the study area with the location of sampling sites. Present day tectonic framework of western Mexico with the relative motions between Rivera and Cocos Plates relative to North America (after *Rosas-Elguera et al., 1996, 2003* and references therein).

Mexican Volcanic Belt) and faults systems which outline main crustal blocks (e.g. Jalisco and Michoacán blocks) (*Rosas-Elguera et al., 2003* and references therein). Paleomagnetic results have largely contributed to understand the tectonics setting of Western and Central Mexico. *Urrutia-Fucugauchi and Bohnel (1987, 1988)* studied the Cretaceous to Quaternary rocks within or immediately to the south of the Trans Mexican Volcanic Belt (TMVB). They reported evidences of several counterclockwise rotations of both local and regional nature.

The recent fault systems in western Mexico are related to extensional processes that formed the Tepic-Zacoalco, Colima, and Chapala grabens (Fig. 1). The motion of the Jalisco and Michoacán blocks occurred along these structures. The motion of the Jalisco block had been interpreted as a crustal block far away to the northwest along the Tepic-Zacoalco graben, as occurred with the peninsula of Baja California since Miocene times (*Luhr et al., 1985*) but structural studies suggest that the motion of the Jalisco block is to the southwest (*Rosas-Elguera et al., 1996*; *Ferrari and Rosas-Elguera, 2000*). On the other hand, according to the paleomagnetic data the motion of the Michoacán block can be reconstructed by considering a SE-motion since Miocene times which can be explained by considering a sinistral oblique subduction of the Cocos Plate relative to North America (*Rosas-Elguera et al., 2003*; *DeMets and Stein, 1990*). It is not clear whether older rocks are affected by similar deformation. Thus, the principal aim of this study is to evaluate whether the tectonic rotations and relative motions of the Michoacán block occurred before the Miocene based on the study of the Eocene andesitic swarm dikes. About 150 standard paleomagnetic cores were collected from 15 (Fig. 1) sites using a pomeroy portable drill. In average, 10 samples were drilled from each site. Samples were oriented in situ by using both, magnetic and solar compasses.

2. GEOLOGICAL SETTING

2.1. The Michoacán Block

The Michoacán block (MB), as the Jalisco block, belongs to the Guerrero Terrene. In this Terrene, the Arteaga Complex underlies a sedimentary, volcano-sedimentary, and volcanic cretaceous-tertiary sequence (*SGM, 2003*). The Arteaga Complex, Triassic in age, is formed by gneiss and schist both intruded by the Tumbiscato meta-intrusive with ages ranging between 127 Ma to 158 Ma (*Grajales-Nishimura and López-Infanzón, 1983*). A cretaceous metavolcano-sedimentary succession that overlies the Arteaga Complex is mantled by the Tecalitlan Formation (~135 Ma) which is a volcanic sequence formed with lava flows, agglomerates, and breccias all of them andesitic in composition. Younger than the Tecalitlan formation are the Tepalcatepec and Madrid formations; first one, is a succession of volcanic rocks at the bottom and sandstone, limonite, shale, arrecifal limestone to the top; whereas the Madrid formation is formed by the Aptian-Albian shale and limestone (*Pimentel, unpublished results*).

The Jilotlan batholith is part of the Mesozoic-Cenozoic magmatic arc assemblage of the central Guerrero Terrane. It is located east to the Colima graben and is part of the plutonic belt along the southern pacific margin of Mexico. At the central part of the Jilotlan batholith the rocks are composed of gabbros (~115 Ma) which are intruded by rocks that range from diorite to granite (~70 Ma average), thus the geochronological data

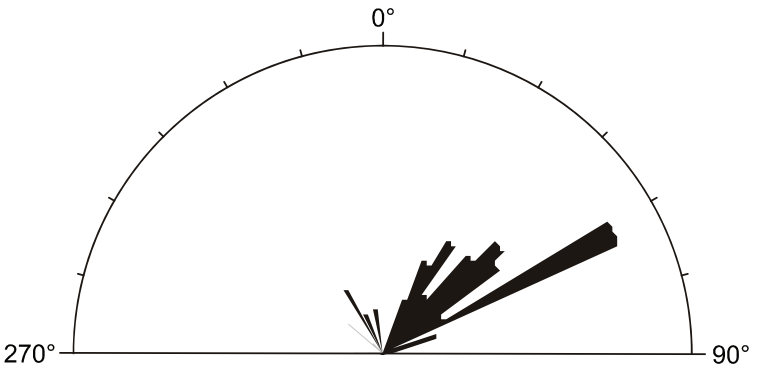


Fig. 2. Preliminary statistical direction of the swarm dike showing a preferential NE-SW trend.

from the Jilotlan batholith suggest two plutonic episodes separated by a ~45 Ma hiatus. (*Schaaf et al., 1995*).

2.2. Dike Swarm

Statistically the dikes swarm can be oriented in a specific direction thus they can be used as tectonic indicators assuming that its direction is normal to the minimal stress field. The Jilotlan batholith, as the cretaceous-tertiary volcano-sedimentary and volcanic successions, is cut by a dyke swarm. There are few roads in this part of the Michoacán block so the ease of access is too much difficult. The sampling was done along the road between Tecalitlan and Tepalcatepec towns. Most of the dikes were highly fracturated which provokes too many troubles for the sampling during the field work. Preliminarily, we mapped at least 106 andesitic dikes. However, consequent field trips revealed some ryholitic and basaltic dikes as well. In average the dikes are 3.5 m thick, with 53° dipping to the NW, and 30 to 250 m long. The mean direction of these dikes is NE-SW (Fig. 2) which agrees with the volcanoes alignments of the southern part of the Michoacán-Guanajuato volcanic field (Fig. 1). We dated a dike giving an age of 35.0 ± 1.8 (see the details below). *Ferrari et al (2002)* reported two magmatic episodes (35–27 Ma and 21–23 Ma) sub-parallel to the present day subduction zone. *Moran-Zenteno et al (2007)* considered three intervals: 68–48 Ma, 48–38 Ma, and 38–28 Ma which were used to outline the general pattern of the distribution of the thermal-magmatic events. In other hand, the geographic distributions of the dated rocks outline magmatic belts sub-parallel to the present day subduction zone. Thus, we believe that the Tecalitlan dike swarm belongs to the Eocene-Oligocene magmatic episode.

2.3. Tectonics According to the Available Paleomagnetic Data

Many paleomagnetic studies along central Mexico have shown that the observed declination do not agree with expected one derived from the North American platform (see reference poles given by *Besse and Courtilot, 2002*). For instance, some late Miocene tectonic events occurred in the Guadalajara triple junction can be explained only if

a rotation of the Jalisco and Michoacán blocks around a vertical axis is considered. During late Miocene, a right-lateral transtension was dominant along the Tepic-Zacoalco graben (*Ferrari, 1995*); although the paleomagnetic studies suggest that no block rotation has been occurred (*Goguitchaichvili et al., 2002*). Also, the Chapala graben was apparently affected by a left lateral transtension (*Garduño et al., 1993*) in agreement with the counterclockwise rotations at Los Altos (*Urrutia-Fucugauchi and Rosas-Elguera, 1994*; *Alva-Valdivia et al., 2000*). On the other hand, some structural events along both, Colima and Chapala grabens (*Barrier et al., 1990*; *Rosas-Elguera et al., 1996*) have been explained based on the sinistral oblique convergence of the Cocos Plate, relative to the North America plate, which induced a southeastward motion of the Michoacán block. The latter was suggested (*DeMets and Stein, 1990*) and confirmed by the paleomagnetic study of Mio-Pliocene rocks from Cotija region. A ~20° counterclockwise rotation of the Michoacán block around a vertical axis during Mio-Pliocene was determined (*Rosas-Elguera et al., 2003*).

3. GEOCHRONOLOGY

Sample TE15 (see Tables 1 and 2), collected from one dike of the swarm, was dated by ^{40}Ar-^{39}Ar analysis at the Geochronology Laboratory of the Geology Department (CICESE). Two experiments were conducted on a groundmass concentrate. A temperature controlled Ta-furnace was used to step-heat the sample and the argon isotopes were analyzed with a MS-10 mass spectrometer connected on-line. The samples were irradiated at position 5C in the U-enriched research reactor of McMaster University for 10 h. As irradiation monitor aliquots of biotite CATAV 7-4, an internal standard, were distributed among the samples, for these the argon isotopes were extracted in one step experiment. Biotite CATAV 7-4 has been calibrated against international reference samples such as hornblende hb 3gr, hornblende MMhb, biotite LP-6 and biotite HD-B1. The argon isotopes were corrected for blank, mass discrimination, radioactive decay of ^{37}Ar and ^{39}Ar and atmospheric contamination. The factors given by *Bottomley and York (1976)* were used to correct for the neutron interference reactions. The decay constants recommended by *Steiger and Jäger (1977)* were applied in the data processing. The equations reported by *York et al. (2004)* were used in all the straight line fitting routines of the argon data reduction. The error in the plateau, integrated and isochron ages, includes the scatter of the irradiation parameter *J*. The analytical precision is reported as one standard deviation (σ). The results obtained on the two experiments performed yielded reproducible results; in Table 1 the relevant argon results are reported. For the second experiment, all the fractions yielded statistically undistinguishable ages within one standard deviation. The plateau age of 35.6 ± 1.6 Ma calculated from the weighted mean of these fractions is in complete agreement with the isochron age calculated with the combined fractions of the two experiments (Fig. 3). We take the isochron age of 35.0 ± 1.8 Ma, as our best estimate for the age of the dike.

Table. 1. Analytical data of $^{40}Ar/^{39}Ar$ (see the text for more details). The site (dike) labeled as TE15 was used for this experiment. T is the temperature reached in the Ta-furnace at which the argon was released; t is age of individual fraction, not including the uncertainty in J; t_i - integrated age; t_p - plateau age calculated with the weighted mean of the fractions selected, these are identified with the symbol §; t_c - isochron age, calculated combining all the fractions of the two experiments performed; all errors are given to 1σ level. *MSWD* is mean square of weighted deviation. Prefered age is higlighted in bold typeface. $J = 0.003647 \pm 0.000021$. The TE15 site (dike) was used for this experiment.

T [°C]	^{39}Ar [%]	$^{40}Ar/^{39}Ar_K$	t [Ma]		^{40}Ar [%]	$^{40}Ar/^{36}Ar$	$^{37}Ar_{Ca}/^{39}Ar_K$
600	0.1077	9.08 ± 2.18	58.8 ± 13.9		4.02	307.88	11.17
800	0.2530	6.49 ± 0.85	42.2 ± 5.5		14.59	345.96	7.63
950	0.2263	6.03 ± 0.93	39.3 ± 6.0		30.95	427.97	5.43
1300	0.4130	5.72 ± 0.52	37.2 ± 3.3		25.64	397.39	19.34
600	0.0849	5.13 ± 1.37	33.4 ± 8.9	§	5.79	313.67	10.85
800	0.1625	5.57 ± 0.67	36.3 ± 4.3	§	19.88	368.83	7.16
900	0.1085	5.51 ± 0.98	35.9 ± 6.3	§	30.65	426.12	4.37
1000	0.1960	5.40 ± 0.55	35.2 ± 3.5	§	34.50	451.12	6.96
1350	0.4481	5.49 ± 0.35	35.8 ± 2.2	§	28.61	413.93	21.61

T [°C]	t_i [Ma]	t_p [Ma]	$(^{39}Ar)_p$	t_c [Ma]	$(^{40}Ar/^{36}Ar)_i$	*MSWD*/*n*
600						
800						
950						
1300	41.3 ±3.0					
600						
800						
900						
1000						
1350	35.5 ±1.9	35.6 ± 1.6	100	**35.0 ± 1.8**	299.6 ± 2.6	0.32/9

4. LABORATORY PROCEDURES

To obtain the low susceptibility measurements (κ-T curves) under air we used the Bartington MS2B susceptibility meter equipped with a furnace. One sample for each site was heated up to 650°C at 20°/min and then cooled at the same rate.

The remanent magnetization was measured with a JR-5A spinner magnetometer (sensitivity $\sim 10^{-9}$ Am^2). Demagnetization was carried out by alternating field demagnetization (AF) and stepwise thermal demagnetization up to 560°C. We used a Molspin AF demagnetizer and a TD-48 furnace to demagnetize 125 samples.

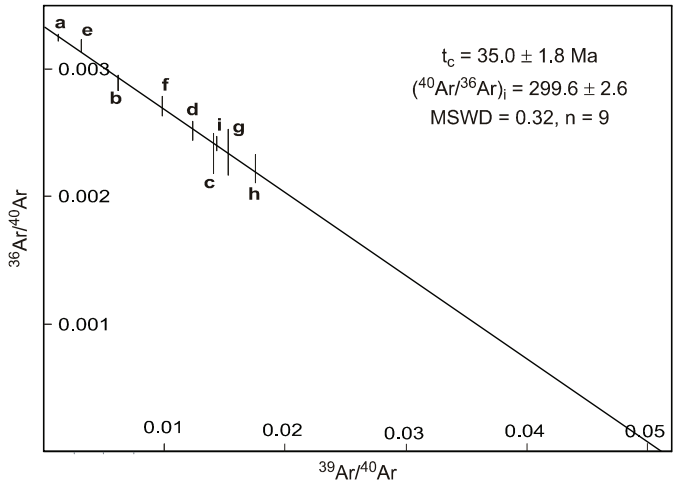


Fig. 3. $^{36}Ar/^{40}Ar$ versus $^{39}Ar/^{40}Ar$ correlation diagram showing the isochron age of 35.0 ± 1.8 Ma, obtained for whole rock Tan 105 sample. The data of all the fractions obtained on the two experiments performed were used (**a** to **d** correspond to the four fractions of the first experiment and **e** to **i** represent the five fractions of the second experiment).

5. ROCK MAGNETIC PROPERTIES

The thermomagnetic curves revealed in most of case, the evidence for Ti-poor titanomagnetites as main magnetic carrier (Fig. 4a). However, several samples showed a more complex behavior during the heating; Several magnetic phases Ti-magnetites and Ti-hematites with variable Ti content may be observed from 550 to 700°C (*Dunlop and Ozdemir, 1997*). Moreover, the cooling and heating curves are not reversible (Fig. 4b). Both alternating field treatments and thermal demagnetization experiments (see below for more details) do not support this hipothesis. The great part of remanence is removed after applying 95 mT peak alternating field or heating at 560°C. It seems that (titano)hematites may coexist in studied samples but their contribution in magnetic remanence is minor.

Hysteresis measurements at room temperature were performed on all samples (one to two samples per dyke) using the AGFM 'Micromag' in fields up to 1 T. The saturation remanent magnetization (J_{rs}), the saturation magnetization (J_s) and coercitive force (H_c) were determined after correction for the paramagnetic contribution. The coercivity of remanence (H_{cr}) was determined by applying progressively increasing backfield after saturation. Typical hysteresis plot is reported reported in Fig. 5a. These curves together with isothermal remanent magnetization (IRM) acquisition experiments (Fig. 5b*)* points to ferrimagnetic phases since the saturation is reached at low to moderate fields (150–200 mT). These results corroborate previous results that infer the presence of Ti-magnetite as the main magnetic carrier. Judging from the ratios of hysteresis parameters (*Day et al., 1977*), it seems that all samples fall in the transition zone between the pseudo-single-domain (PSD) and single-domain (SD) grain size regions (Fig. 5c).

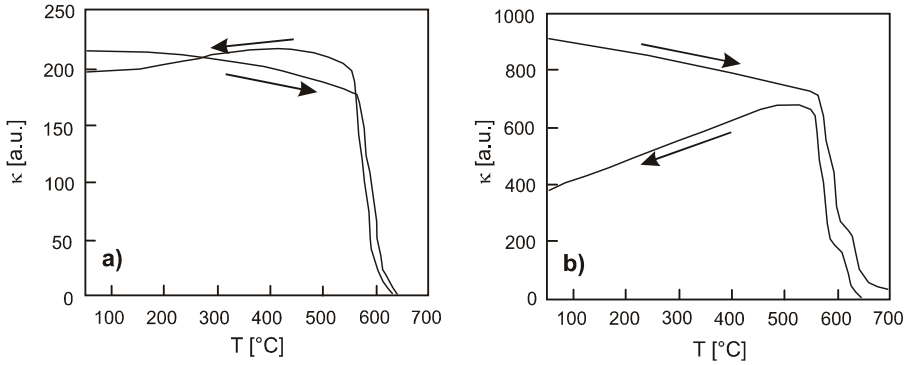


Fig. 4. Susceptibility κ versus temperature T (in air) curves of representative samples at high temperatures. **a)** low-Ti titanomagnetite, and **b)** low-Ti titanomagnetite with probably titanohematite.

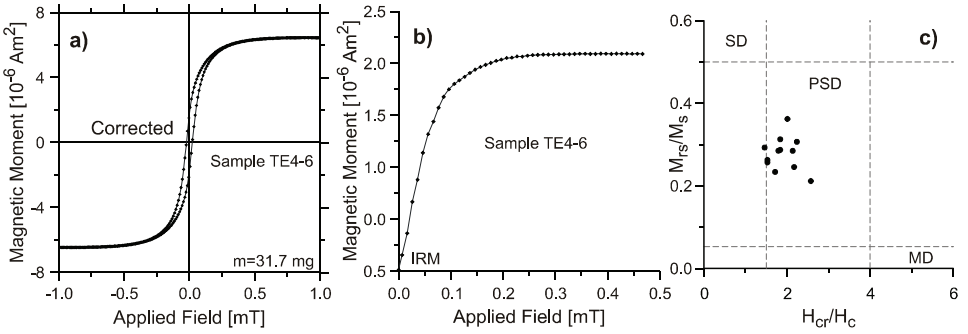


Fig. 5. **a)** Representative hystresis plot (corrected for dia- and paramagnetism), and **b)** associated isothermal remanence acquisition curve; **c)** Day diagram (*Day et al., 1977*).

Stable paleomagnetic components may be recognized for most of samples (Fig. 6). In cases of thermal demagnetization, the greater part of the remanent magnetization was removed at temperatures between 500 and 550°C (Fig. 6a), which indicates Ti-poor titanomagnetite as responsible for the remanence. A minor secondary component, probably of viscous origin, was easily removed by applying a field between 10 and 20 mT (Fig. 6b). Medium destructive fields range from 35 to 40 mT, which indicates small pseudo-single-domain magnetic grains.

The characteristic magnetization direction was determined by the least squares method (*Kirschvink, 1980*) by taking four to seven points in the principal component analyses.

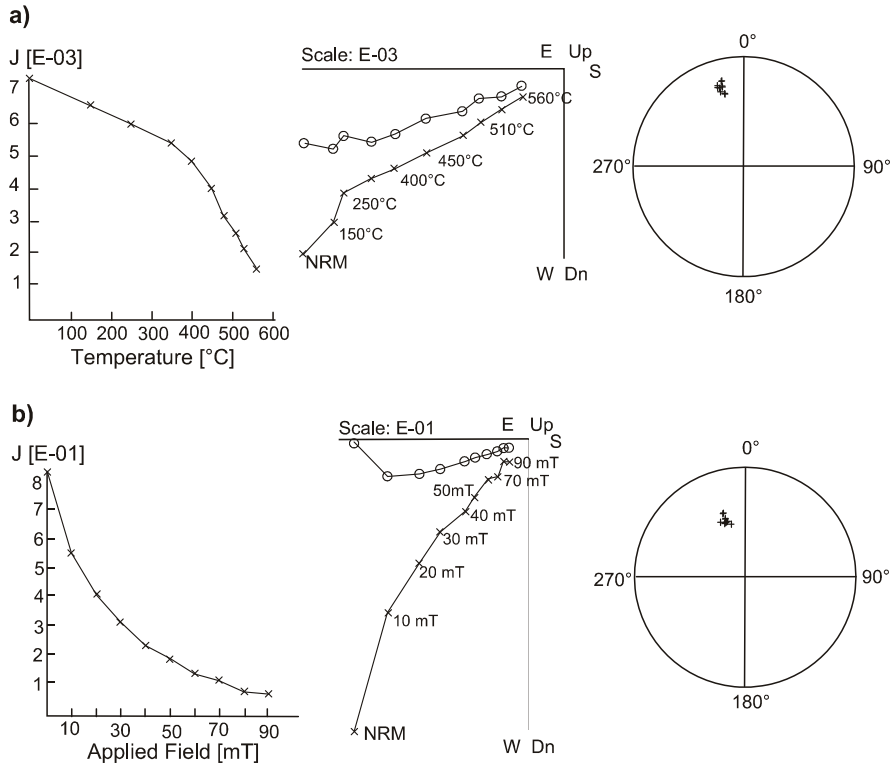


Fig. 6. Orthogonal vector plots: **a)** stepwise thermal and **b)** alternating field demagnetization (stereographic projection) of representative samples.

6. PALEOMAGNETIC DATA

The obtained directions were averaged by site, and the statistical parameters were calculated assuming a Fisherian distribution (Fig. 7). Table 2 shows the site mean directions of cleaned remanence. The majority of sites (11) are characterized by normal polarity magnetizations. In Fig. 7, the mean paleomagnetic direction was computed with the Cotija Mean paleodirection and the expected mean paleodirections for Eocene and Miocene periods retrieved from the reference poles of *Besse and Courtillot (2002)*.

7. MAIN RESULTS AND DISCUSSION

Rock-magnetic experiments permitted identification of magnetic carriers and assessment of the paleomagnetic stability. Continuous susceptibility measurements vs. temperature, in most cases yields reasonably reversible curves with Curie points close to that of magnetite. However, the presence of a minor magnetic phase with Curie

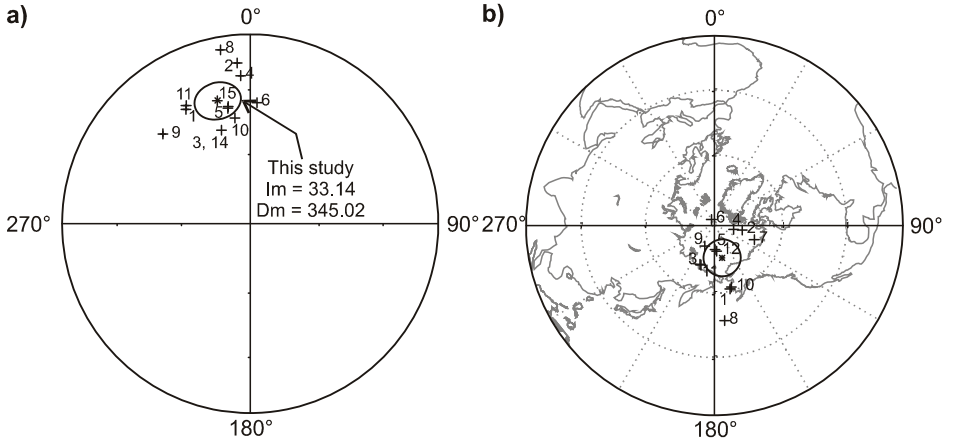


Fig. 7. **a)** Equal area projections of the flow-mean characteristic paleodirections for the Tecalitlan dikes. The numbers represent the studied sites. The mean paleodirection is shown with respective α_{95}. **b)** Site mean virtual geomagnetic pole positions. Same notations as in a).

temperatures above 600°C is also detected in almost all samples. However, the hysteresis and IRM measurements do not support this hypothesis. Although there are some evidences for the presence of titanohematite phase, the remanence seems to be dominated by titanomagnetites (almost magnetite phase). There is also a remote possibility that titanohematites are formed during the laboratory heating in air above 580°C. The mean paleomagnetic direction obtained in this study is $I = 33.1°$, $D = 345.0°$, $k = 25$, $\alpha_{95} = 8.9°$ and the corresponding mean paleomagnetic pole position is $P_{lat} = 75.7°$, $P_{long} = 166.6°$, $K = 31$, $A_{95} = 8.0°$ (Fig. 7a,b). The mean inclination is in reasonably good agreement with the expected value, as derived from reference poles for the stable North America (*Besse and Courtilot, 2002*). Magnetic declination is not significantly different (rotation parameter $R = -6°$ with $\Delta R = 9.2°$) from that expected which is in disagreement with a counterclockwise tectonic rotation of about 20° previously reported for the studied area (Fig. 8). The Cotija Miocene volcanic field studied by *Rosas-Elguera et al. (2003)* yielded $R = -17.6°$ with $\Delta R = 5.5°$).

The W-SW of Mexico has been affected by the subduction of the Farallon plate beneath North America generating smaller plates (e.g. Cocos Plate, *Atwater, 1970*), uplift of crustal blocks with fault system outlining them (e.g. Jalisco and Michoacán blocks) (*Rosas-Elguera et al., 1996*; *Righter et al., 2010*) and magmatic arcs (e.g. Paleocene-Early Oligocene magmatism and the Trans-Mexican Volcanic Belt, *Ferrari et al., 2002*; *Moran-Zenteno et al., 2007*). Furthermore, paleomagnetic results have shown a disagreement between the observed and expected declination, including Cretaceous to Quaternary rocks, suggesting that counter-clockwise block rotations in central Mexico is related to a sinistral fault system (*Urrutia-Fucugaunchi and Bohnel, 1988*; *Alva-Valdivia et al., 2000*).

The new Ar-Ar radiometric age obtained for the Tecalitlan dike swarm agree with the ages of the Paleocene-Early Oligocene magmatism previously reported (e.g *Moran-*

Table 2. Flow-mean paleodirections of characteristic remanence, location. N - number of treated samples; n - number of specimens used for calculation; I - inclination; D - declination; k and α_{95} are precision parameter and radius of 95% confidence angle of the Fisher statistics.

Site	Long [°W]	Lat [°N]	n/N	I_M [°]	D_M [°]	α_{95} [°]	k	P_{lat} [°]	P_{long} [°]
TE01	103.26	19.46	5/6	31.22	330.40	7.9	72	61.79	166.28
TE02	103.26	19.45	5/6	16.03	355.28	7.8	75	77.83	99.47
TE03	103.26	19.45	4/5	47.02	342.79	14.5	74	72.01	199.15
TE04	103.26	19.45	4/5	22.67	356.41	14.8	39	81.60	101.62
TE05	103.16	19.43	6/7	37.91	349.13	8.9	58	79.65	178.97
TE06	103.01	19.36	4/6	36.09	2.97	14.8	40	87.13	333.11
TE08	103.01	19.35	5/6	7.55	350.27	10.8	51	71.77	109.61
TE09	102.85	19.23	7/7	34.03	315.66	10.5	34	48.17	173.94
TE10	102.86	19.23	5/6	43.16	351.50	12.2	40	80.17	205.5
TE11	102.86	19.23	6/6	29.59	331.30	12.8	28	62.46	164.66
TE14	101.92	19.18	6/7	47.10	343.22	11.3	36	72.24	200.64
TE15	101.92	19.18	7/7	36.80	348.30	7.8	57	78.94	175.34
			Mean	33.14	345.02	8.9	25	75.71	166.60

Zanteno et al., 2007). Considering the regional tectonic framework however, the dike swarm with a mean NE-SW-trend (Fig. 2) and an overall mean paleodirection of $I = 33.1°$, $D = 345.0°$, $\alpha_{95} = 8.9°$ (Fig. 8) has a main trouble. The declination is in agreement with the expected for Eocene time nevertheless younger rocks of the Michoacán block have shown counterclock rotation at the Cotija area (Fig. 8).

The subduction of the oceanic plate beneath a continental plate caused both deformation of the overriding plate and intense volcanic activity. In this context, several aspects may be involved as the convergence vector relative to the trend of the trench at the subduction zone, convergence rate itself and deformation of older structures. All these factors contribute to define the style of intra-plate deformation. The simplest tectonic models reported are based on relationship between the extensional tectonics to doming and subsequent rifting (*Closs, 1939*). The experiments performed by Closs show that tensile fractures were formed after of doming processes and are parallel to the maximum horizontal stress (σ_1) direction. This simple model may explain the paleodirections obtained for the the dike swarm in the Michoacán block assuming that sometime in Eocene epoch the convergence vector (in our case with NE-trend) was normal to the trench (Fig. 9a) before reaching an actual oblique convergence of the Farallon plate relative to North America. In this new tectonic framework a left-lateral transpression was formed in the overriding plate forming fractures subparallel to the trench (Fig. 9b). Similar structure may be observed in Chapala-Oaxaca fault zone (*Harrison and Johnson, 1988*) and the Apatzingan fractures (*Garduño et al., 1999*) and seems to be a general characteristic of the region.

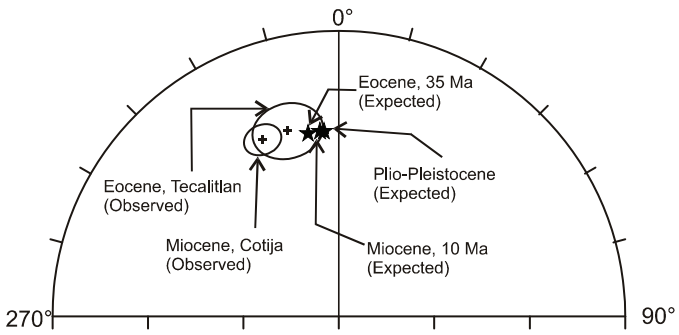


Fig. 8. Equal area projection of the overall mean Paleodirection for Eocene Tecalitlan (this study) and the Miocene Cotija (*Rosas Elguera et al., 2003*) volcanic fields. The expected paleodirections for Eocene, Miocene and Plio-Pleistocene periods (*Besse and Courtillot, 2002*) are also shown.

8. CONCLUSIONS

The oblique subduction of the Farallon plate beneath American plate left behind smaller plates (e.g Cocos plate), fault systems outlining crustal blocks (e.g. Michoacan block). Our detailed rock-magnetic and paleomagnetic results from the Eocene (35 Ma) dike swarm (with N-NE trend) from Tecalitlan area, located in the Michoacan block suggest that the mean inclination agree with the expected value, as derived from reference poles for the stable North America. Magnetic declination is not significantly different from that expected which is in disagreement with a counterclockwise tectonic rotation of about 20º previously reported for the studied area. With these results, and the compiled from the Michoacan Block, we propose a simple tectonic model suggesting that sometime in Eocene epoch the convergence vector of the Farallon plate relative to North America plate was normal to the trench before reaching an actual oblique convergence.

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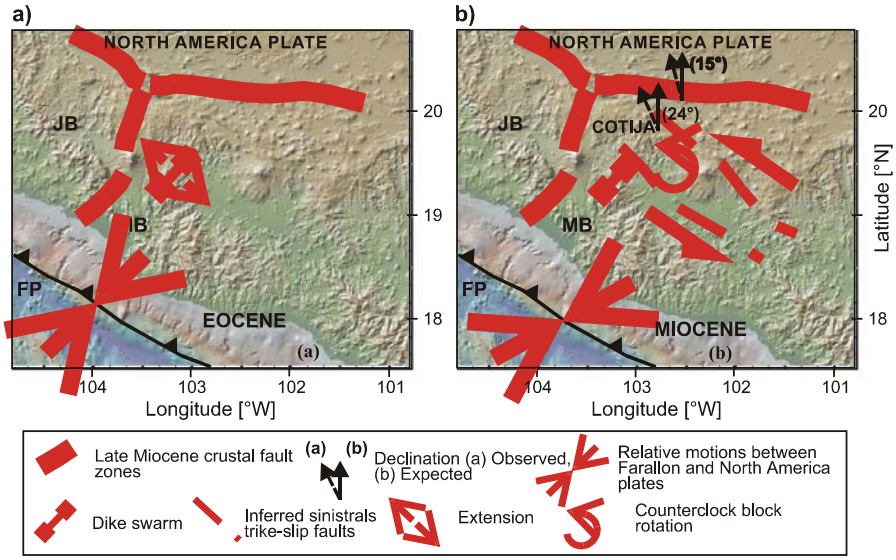


Fig. 9. Tectonic setting for Eo-Miocene periods according to: **a)** present field, and **b)** paleomagnetic data (see text for more details).

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